MILITARY SPECIFICATION

WARHEAD, GM, HE, XM248E1
INERT PARTS FOR

This specification is approved for use by the U.S. Army Armament Research and Development Command, and is available for all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This document contains the quality assurance provisions and special requirements for the fabrication of the inert parts for one type of warhead designated as Warhead, GM, HE, XM248E1.

2. APPLICABLE DOCUMENTS

2.1 Issues of Documents - The following documents of the issue in effect on date of invitation for bids form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-C-6021	- Castings, Classification and Inspection Of
MIL-I-6866	- Inspection, Penetrant, Method of
MIL-A-21180	- Aluminum Alloy Castings, High Strength
MIL-W-22248	- Weldments, Aluminum and Aluminum Alloy
MIL-L-25567	- Leak Test Compound, Oxygen Systems
MIL-I-45607	- Inspection Equipment; Acquisition, Maintenance and Disposition of
MIL-A-48078	- Ammunition Standard Quality Assurance Provisions, General Specificatons For

FSC: 1336

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

MIL-STD-453 - Inspection Radiographic
MIL-STD-1169 - Packaging, Packing and Marking For Shipment of Inert Ammunition Components

DRAWINGS (See 6.4)

US ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND

9312581 - Warhead XM248E1 Inert Parts Assembly
9282798 - Packing and Marking for Box, Wood, for Warhead GM, HE, XM248E1 or XM248E1 Inert Parts Assembly

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functons should be obtained from the procuring activity or as directed by the Contracting Officer).

2.2 Other Publications - The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM B247 -Aluminum Alloy Die and Hand Forgings

(Application for copies shall be addressed to the Americn Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103).

3. REQUIREMENTS

3.1 Materials - Materials used in fabrication of the Warhead shall be in accordance with the applicable drawings and specifications.

3.1.1 Castings - Castings shall comply with the requirements for class and grade as stated on the applicable drawing and in the applicable specifications.

3.1.2 Forgings - Forgings shall be in accordance with the applicable drawing and shall comply with the requirements of ASTM

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B247 - Aluminum Alloy Die and Hand Forgings. A certificate of compliance shall be furnished by the producer for each lot of forgings.

3.2 Components and Assemblies - The components and assemblies shall comply with all the requirements specified on drawing 9312581 and associated drawings, and with all the requirements specified in applicable specifications and standards.

3.3 Air Test Certification of Welds - The welds of the Metal Parts Welded Assembly shall withstand an internal air pressure of 10 psi for a minimum of twenty (20) seconds with no evidence of leakage.

3.4 Adhesive Bonding Procedures - The contractor shall establish a standard operating procedure (SOP) for the adhesive bonding of the inert parts assembly, dwg. 9312581, and shall submit this SOP for approval following the same instructions set forth herein for the submission of inspection equipment designs. Changes to the approved SOP will require the approval of the Contracting Officer.

3.5 First Article Inspection - This specification makes provisions for First Article Inspection. Requirements for submission of First article Samples (FAS) by the contractor shall be as specified in the contract.

3.6 Workmanship - All parts shall be fabricated and finished in a thorough, workmanlike manner. They shall be free of burrs, chips, sharp edges, cracks, unblended radii, surface defects, dirt, grease, corrosion products and other foreign matter, Required markings shall be neat and sharply defined. Cleaning methods used shall not be injurious to any part, nor shall the parts be contaminated by the cleaning agent. All helicoil inserts shall be installed below flush and their tangs removed, similarly the lock rings for the studs shall be seated flush or below. Attentive care shall be taken during fabrication, inspection and handling to prevent nicking and scoring of the aluminum base rim and end. In fragment lay up the base rim shall be kept clean of adhesives. The layup shall be free from blisters and fragment vacancies, and the spacing between fragments and fragment rows shall be uniform over each assembly. Any gaps between the outer wrap and the nose lip shall be filled in with adhesive.

3.6.1 Workmanship of Weldments - Weldments shall be free from slag, flux, salt baths residue, weld spatter and other foreign

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materials and shall meet the porosity standards of the drawing and any applicable specification identified on the drawing.

3.6.2 Workmanship of Castings - Castings shall comply with the workmanship requirements of MIL-A-21180, Aluminum Alloy Castings, High Strength.

3.6.3 Workmanship of Forgings - Forgings shall be of uniform quality and free from blisters, fins, seams, laps, bursts, cracks, segregations or other injurious defects.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection and Standard Quality Assurance Provisions - Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this specification.

4.2 Classification of Inspection - The following types of inspection shall be conducted on this item:

a. First Article Inspection
b. Quality Conformance Inspection

4.3 First Article Inspection

4.3.1 Submission - The contractor shall submit a First Article Sample as designated by the Contracting Officer for evaluation in accordance with the provisions of 4.3.2. The First Article Sample shall consist of the following items in sample quantities as indicated:

Part Description	Drawings	Quantity
Retaining Cup	9322123	5
Ring	9312612	5
Retaining Cup/Ring Assembly	9312584	5
Nose Plate	9312518	5
Nose	9312565	5
Shell	9312566	5
Base	9278872	5
	if casting	also 5 castings per 4.3.1.1
	if forging	also 5 forgings per 4.3.1.2

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Metal Parts Welded Assembly	9312580 if e.b. welding	5 also 6 weld specimens per 4.3.1.3
Fragment	9226307	20
Cover, Base	9328026	5
Gasket	9326707	5
Warhead, XM248E1 Inert Parts Assembly	9312581	5 Plus an approved SOP for Bonding

4.3.1.1 First Article Sample Submission of Base Castings - Where the contractor intends to utilize castings for manufacture of the base (dwg. 9278872), the contractor shall submit a First Article Sample consisting of five (5) unmachined base castings for evaluation in accordance with the provisions of 4.3.2.

4.3.1.2 First Article Sample Submission of Forgings - Where the contractor intends to utilize forgings for manufacture of the base (dwg. 9278872), the contractor shall submit a Frist Article Sample consisting of five (5) unmachined forgings for evaluation in accordance with the provisions of 4.3.2.

4.3.1.3 First Article Sample Submission if Electron Beam Welding - Where the contractor intends to utilize electron beam (e.b.) welding for manufacture of the metal parts welded assembly, the contractor shall submit six e.b. weld specimens (see paragraph 4.5.4) in addition to the five (FAS) required metal parts welded assemblies (9312580). Five of these six specimens shall be welded prior to the (FAS) assemblies, and the remaining specimen shall be welded after the assemblies.

4.3.2 Inspections to be Performed - See MIL-A-48078 and Table I specified herein.

4.3.2.1 Inspections to be performed if Electron Beam Welding - Each of the six e.b. weld specimens (see 4.3.1.3) and two of the five required assemblies shall be sectioned and one surface polished and metallographically examined to determine compliance to the drawing requirements for the base-shell weldments. Sectioning of the two metal parts welded assemblies shall be accomplished upon completion of all the other required First Article Inspections (see Table I).

4.3.3 Rejection - See MIL-A-48078.

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TABLE 1 - FIRST ARTICLE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-W-63328(AR)

PARAGRAPH 4.3.3.1	TITLE		SHEET 1 OF 3		DRAWING NUMBER See below
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Retaining Cup (Dwg. 9322123) Examination for Defects	5	-	3.2	4.4.2.1
	Ring (Dwg. 9312612) Examination for Defects	5	-	3.2	4.4.2.2
	Retaining Cup/Ring Assembly (Dwg. 9312584) Examination for Defects	5	-	3.2	4.4.2.3
	Nose Plate (Dwg. 9312518) Examination for Defects	5	-	3.2	4.4.2.4
	Nose (Dwg. 9312565) Examination for Defects	5	-	3.2	4.4.2.5
	Shell (Dwg. 9312566) Examination for Defects	5	-	3.2	4.4.2.6
	Examination for Hardness	5	-	3.2	Note (a)

NOTES: (a) As part of the First Article Inspection only, the temper condition of each shell in the (FAS) shall be verified by a hardness test. Shells shall exhibit a minimum hardness of 90 Rockwell E, or the equivalent BHN or Rockwell B.

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TABLE 1 - FIRST ARTICLE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS MIL-W-63328(AR)

PARAGRAPH	TITLE		SHEET 2 OF 3		DRAWING NUMBER
4.3.3.1 (Cont'd)					See below
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Base (Dwg. 9278872) Examination for Defects	5	-	3.1,3.2 3.6	4.4.2.7
	Base Castings Examination for Defects	5	-	3.1,3.2 3.6	Note (b)
	Base Forgings Examination for Defects	5	-	3.1,3.2 3.6	Note (c)

(b) Each casting shall be examined for casting discontinuities (see para. 4.5.1); and two (2) castings shall be examined for mechanical properties as stated in 4.5.2, on the drawing, and in the applicable specifications.

(c) Each forging shall be visually examined for surface defects (see para. 3.6 and ASTM B247). Confirmation of heat treat response shall be established by examining two (2) PAS forgings for mechanical properties as stated in ASTM-B247 (one tensile specimen shall be taken from the rim and one from the floor of each forging).

NOTES:

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TABLE I - FIRST ARTICLE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-W-63328 (AR)

PARAGRAPH	TITLE		SHEET 3 OF 3		DRAWING NUMBER
4.3.3.1 (Cont'd)					See below
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Metal Parts Welded Assembly (Dwg. 9312580)				
	Examination for Defects	5	-	3.2,3.3 3.6	4.4.2.8, 4.3.2.1
	Fragment (Dwg. 9226307)				
	Examination for Defects	20	-	3.2	4.4.2.9
	Cover, Base (Dwg. 9328026)				
	Examination for Defects	5	-	3.2	4.4.2.10 and note (d)
	Fit check				
	Wrap, Outer (Dwg. 9312518)				
	Certificate of conformance	-	-	3.1	-
	Gasket (Dwg. 9326707)				
	Examination for Defects	5	-	3.2	4.4.2.11
	XM248E1 Inert Parts Assembly (Dwg. 9312581)				
	Examination for Defects	5	-	3.2,3.6	4.4.2.12

NOTES: (d) Each base cover of the FAS shall be fitted over the base of a metal parts welded assembly to assure the adequacy of fit and the proper location of the small holes in the cover with the large insert holes of the metal parts welded assembly.

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4.4 Quality Conformance Inpsection

4.4.1 Inspection Lot Formation - Inspection lots shall comply with the lot formation provisions of MIL-A-48078.

4.4.2 Examination - See MIL-A-48078.

a. Sampling Plans - Unless otherwise specified in the Classification of Defects and Test tables, sampling plans for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-W-63328 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9322123
4.4.2.1	Retaining Cup				NEXT HIGHER ASSEMBLY 9312584
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Outside diameter		1.0%	3.2	Gage
102	Wall thickness		1.0%	3.2	Gage
103	Corner Radius		1.0%	3.2	Gage
104	Certification of material missing		Note (a)	3.1	Visual
105	Length		0.4%	3.2	Gage
Minor					
201	Evidence of poor workmanship		1.0%	3.5	Visual

NOTES:

(a) Verification shall be made that the material stock conforms to its applicable requirements. A certificate of conformance from the supplier shall be available.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-W-63328 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9312612
4.4.2.2	Ring				NEXT HIGHER ASSEMBLY 9312584
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Inside diameter		0.40%	3.2	Gage
102	Length from large outside diameter to aft end		0.40%	3.2	Gage
103	Counter bore diameter		0.40%	3.2	Gage
104	Depth of groove		0.40%	3.2	Gage
105	Distance from aft end to start of groove		0.40%	3.2	Gage
106	Length of groove diameter		0.40%	3.2	Gage
107	Certification of material missing		Note (a)	3.1	Visual
Minor					
201	Large outside diameter		0.65%	3.2	Gage
202	Evidence of poor workmanship		1.0%	3.5	Visual

NOTES: (a) Verification shall be made that the material stock conforms to its applicable requirements. A certificate of conformance from the supplier shall be available.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET OF		DRAWING NUMBER
4.4.2.3	Retaining Cup/Ring Assembly		1 1		9312584
					NEXT HIGHER ASSEMBLY 9312580
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Penetrant examination of weld		100%	3.2	See Note below/Visual
102	Air test		100%	3.2	Gage
103	Distance from cup bottom to ring shoulder		0.40%	3.2	Gage
104	Inside diameter, min.		0.40%	3.2	Gage
Minor					
201	Evidence of poor workmanship		1.0%	3.5	Visual

NOTE: Each Retaining Cup/Ring Assembly weldment shall be penetrant examined in accordance with MIL-I-6866 (Type I, Method A or B) and shall comply with the requirements set forth in MIL-W-22248 for Class III weldments, or when electron beam welding, to the requirements stated on the drawing. Any assembly which fails to comply, shall be rejected.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-W-63328 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9312518
4.4.2.4	Nose Plate				NEXT HIGHER ASSEMBLY 9312581
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	True position of holes to outside diameter		0.40%	3.2	Gage
102	Distance from bottom of hole countersink to end		0.40%	3.2	Gage
103	Runout of face seat to outside diameter		0.40%	3.2	Gage
104	Length of large outside diameter		0.40%	3.2	Gage
105	Certification of material missing		Note (a)	3.1	Visual
Minor					
201	Small outside diameter		1.0%	3.2	Gage
202	Length overall		1.0%	3.2	Gage
203	Diameter of holes, min.		1.0%	3.2	Gage
204	Countersink diameter, min.		1.0%	3.2	Gage
205	Protective finish missing or improper		1.0%	3.2	Visual
206	Evidence of poor workmanship		1.0%	3.5	Visual

NOTES: (a) A certificate of conformance from the material stock supplier shall be available for verification by the inspector.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-W-63328 (AR)

PARAGRAPH	TITLE			SHEET OF 1 1	DRAWING NUMBER 9312565
4.4.2.5	Nose				NEXT HIGHER ASSEMBLY 9312580
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	True position of holes to inside diameter		0.40%	3.2	Gage
102	Inside diameter		0.40%	3.2	Gage
103	Length of inside diameter		0.40%	3.2	Gage
104	Penetrant examination for casting discontinuities	see note	MIL-C-6021	3.1.1	See Note Below
105	Outside diameter of forward end		0.40%	3.2	Gage
106	Pitch diameter of threaded hole (prior to helicoil installation)		0.65%	3.2	Gage
107	Minor diameter max, of threaded hole (prior to helicoil installation)		0.65%	3.2	Gage
Minor					
201	Radius of surface A at point A		1.0%	3.2	Gage
202	Radius of surface A at point B		1.0%	3.2	Gage
203	Depth of step in plane AC BD		1.0%	3.2	Gage
204	Evidence of poor workmanship		1.0%	3.5	Visual

NOTES: The nose if cast, shall be penetrant examined in accordance with MIL-I-6866 (Type 1, Method A) using the sampling plan and acceptance criteria for class 2B castings specified in MIL-C-6021 Table 1. See paragraph 4.3 of MIL-C-6021 for rejection criteria. Inspection applies for cast noses only.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-W-63328 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9312566
4.4.2.6	Shell				NEXT HIGHER ASSEMBLY 9312580
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Profile		0.40%	3.2	Gage
102	Abrupt change in contour		0.40%	3.2	Gage
103	Wall thickness between surface A and max. dia.		0.40%	3.2	Gage
104	Wall thickness between surface B and max. dia.		0.40%	3.2	Gage
105	Diameter at widest point		0.40%	3.2	Gage
106	Distance from base end to max. diameter		0.40%	3.2	Gage
107	Certification of material missing	by lot	Note (a)	3.1	Visual
Minor					
201	Length, min.		0.65%	3.2	Gage
202	Outside diameter of nose		0.65%	3.2	Gage
203	Inside diameter of nose		0.65%	3.2	Gage
204	Inside diameter of base end		0.65%	3.2	Gage
205	Wall thickness variation		0.65%	3.2	Gage
206	Chamfer missing at base end		0.65%	3.2	Visual/See note (b)
207	Evidence of poor workmanship		1.0%	3.5	Visual

NOTES:
(a) A certificate of conformance from the material stock supplier shall be available for verification by the inspector. It shall attest to the T6 condition of the finished item.
(b) Inspection does not apply where contractor elects to electron beam weld at assembly 9312580.

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PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9278872
4.4.2.7	Base				NEXT HIGHER ASSEMBLY 9312580
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Examination for casting discontinuities per MIL-C-6021	-	100% Note (a)	3.1.1	4.5.1/Radiograph and Penetrant
102	Examination for mechanical properties per MIL-A-21180		Notes (a) & (b)	3.1.1	4.5.2/Gage
103	Thickness of base floor		0.40%	3.2	Gage
104	Width of depressions in base side, min.		100%	3.2	Gage
105	Distance from center to bottom of depression in base side		100%	3.2	Gage
106	Radius missing or improper in and about the base side depressions		100%	3.2	Visual
107	Distance from end to counterbored surface at center hole		0.40%	3.2	Gage
Minor					
201	Diameter of center hole		0.65%	3.2	Gage
202	True position of round teat to center hole, and end		0.65%	3.2	Gage
203	Diameter of teat, max.		0.65%	3.2	Gage
204	Casting or heat treat numbers missing or improper		1.0%	3.2	Visual, Note (a)
205	Evidence of poor workmanship		1.0%	3.6	Visual
	(a) Examinations 101, 102, and 204 apply to cast bases only.				
NOTES:	(b) One base casting from each heat treat shall be destructively examined for mechanical properties.				

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CLASSIFICATION OF DEFECTS & TESTS

MIL-W-63328(AR)

PARAGRAPH	TITLE		SHEET 1 OF 3		DRAWING NUMBER 9312580
4.4.2.8	Metal Parts Welded Assembly				NEXT HIGHER ASSEMBLY 9312581
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Base diameter		100%	3.2	Gage
102	Runout of base diameter to shell neck diameter and center hole		100%	3.2	Gage
103	Runout of base end to shell neck diameter and center hole		100%	3.2	Gage
104	Flatness of base end		100%	3.2	Gage
105	Distance from center to spot face on rim		0.40%	3.2	Gage
106	Diameter of radial holes in base rim		0.40%	3.2	Gage
107	Location of radial holes in base rim		100%	3.2	Gage
108	True position of four small insert holes to base diameter and key slot (prior to helicoil installation)		100%	3.2	Gage
109	True position of untapped holes to base diameter and key slot		100%	3.2	Gage
110	Diameter of untapped holes		100%	3.2	Gage
111	Depth of untapped holes		100%	3.2	Gage
112	Penetrant examination of welds		100%	3.2	Visual
113	Weld above nose face		0.40%	3.2	Visual
114	Weld above rectangular recess		0.40%	3.2	Visual
115	Interior examination of base-shell weld penetration (heliarc)		100%	3.2	Visual/Note (b)
	(a) The welds of each metal parts assembly shall be penetrant examined in accordance with MIL-I-6866 (Type I Method A or B). Heliarc welds shall comply with the requirements set forth in MIL-W-22248 for Class III weldments; electron beam welds shall comply with the applicable drawing requirement.				
NOTES:	(b) This inspection applies to heliarc welding only.				

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PARAGRAPH	TITLE		SHEET 2 OF 3		DRAWING NUMBER 9312580
4.4.2.8 (Cont'd)	Metal Parts Welded Assembly				NEXT HIGHER ASSEMBLY 9312581
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Major 116	Metallographic examination of base-shell weld specimen (electron-beam)	2 per shift per machine	-	3.2	4.5.4/Gage/Note (c)
117	Air test of welded assembly		100%	3.3	4.5.3/Gage
118	Length of assembly, max.		100%	3.2	Gage
119	True position of studs to base diameter key slot and rectangular recess		100%	3.2	Gage
120	True position of eight small insert holes to base diameter, end and key slot (prior to heli-coil installation)		100%	3.2	Gage
121	True position of four large insert holes to base diameter, end and key slot (prior to helicoil installation)		100%	3.2	Gage
122	Depth of four large insert holes		100%	3.2	Gage
123	Depth of small insert holes @ 4:00 and 6:00		100%	3.2	Gage
124	True position of slot to base diameter, rib, and end		100%	3.2	Gage
125	Width of slot		100%	3.2	Gage
126	Depth of slot		100%	3.2	Gage
127	Flatness of rectangular recess		0.40%	3.2	Gage
128	Length of rectangular recess		0.40%	3.2	Gage
129	Distance from base end to rectangular recess		0.40%	3.2	Gage
130	Pitch diameter of small base holes (prior to helicoil installation)		0.65%	3.2	Gage

NOTES:
(c) This inspection applies to electron-beam welding only.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 3 OF 3		DRAWING NUMBER 9312580
4.4.2.8 (Cont'd)	Metal Parts Welded Assembly				NEXT HIGHER ASSEMBLY 9312581
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Major					
131	Minor diameter max, of small base holes (prior to helicoil installation)		0.65%	3.2	Gage
132	Flatness of nose		0.40%	3.2	Gage
Minor					
201	Width of spot face		1.0%	3.2	Gage
202	Length of spot face		1.0%	3.2	Gage
203	Pitch diameter of four large base holes (prior to helicoil installation)		1.0%	3.2	Gage
204	Minor diameter max, of four large base holes (prior to helicoil installation)		1.0%	3.2	Gage
205	Inserts above flush, not clean or tangs not removed		1.0%	3.6	Visual
206	Stud threads damaged or lock-rings for studs above flush		0.65%	3.6	Visual
207	Serialization missing or improper		0.65%	3.2	Visual
208	Protective finish missing or improper		1.0%	3.2	Visual
209	Evidence of poor workmanship		1.0%	3.6	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.9	Fragment				9226307
					NEXT HIGHER ASSEMBLY 9312581
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Length	*	2.5%	3.2	Gage
102	Height	*	2.5%	3.2	Gage
Minor					
201	Fragment no clean, oily or rusted	*	4.0%	3.2	Visual
202	Presence of burr	*	4.0%	3.2	Visual

NOTES: *Sampling shall be in accordance with special Inspection Level S-3 of MIL-STD-105. An inspection lot shall consist of 10,000 to 35,000 fragments.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS MIL-W-63328 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9328026
4.4.2.10	Cover, Base				NEXT HIGHER ASSEMBLY 9312581
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Inside diameter		4.0%	3.2	Gage
102	Evidence of cracks or tears		4.0%	3.2	Visual
Minor					
201	Small holes missing		6.5%	3.2	Visual
202	Evidence of sharp edges or unremoved flashing		6.5%	3.6	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-W-63328 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.11	Gasket				9326707
					NEXT HIGHER ASSEMBLY 9312581
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	True position of holes		4.0%	3.2	Gage
102	Certification of material		-	3.1	Visual
Minor					
201	Outside diameter, max.		6.5%	3.2	Gage
202	Inside diameter, min.		6.5%	3.2	Gage
203	Thickness, min.		6.5%	3.2	Gage
204	Evidence of damage		6.5%	3.6	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.12	Warhead XM248E1 Inert Parts Assembly				9312581
					NEXT HIGHER ASSEMBLY 9282798
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Weight		100%	3.2	Gage
102	Verification of bonding procedures		100%	3.2	See Note Below
103	Profile		100%	3.2	Gage
Minor					
201	Nose plate missing		1.0%	3.2	Visual
202	Screw or gasket missing		1.0%	3.2	Visual
203	Base cover missing		1.0%	3.2	Visual
204	Evidence of poor workmanship		0.65%	3.6	Visual

NOTES: An inspector shall verify that the warhead shell and fragments have been cleaned/degreased prior to frag. layup, that the required cure times and temperatures have been maintained, and that the shelf-life of the adhesives has not been exceeded.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.13	Packing and Marking for Box, Wood, for Warhead GM HE XM248E1 or XM248E1 Inert Parts Assembly	1 OF 1	9282798
			NEXT HIGHER ASSEMBLY: --------

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Cushion or spacer missing or improper		1.0%	3.2	Visual
102	Strapping missing or improper		1.0%	3.2	Visual
Minor					
201	Marking missing or improper		1.5%	3.2	Visual
202	Base cover missing		1.5%	3.2	Visual
203	Evidence of poor workmanship		1.5%	3.2	Visual

NOTES:

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4.4.3 Testing - Testing is described in the Quality Conformance Inspection tables.

4.4.4 Inspection Equipment - The inspection equipment required to perform the examination and tests prescribed in this specification is identified, either directly or by reference, in the "Paragraph Reference/Inspection Method" column of the Quality Conformance Inspection Tables herein. The contractor shall submit inspection equipment designs to the Government for approval in accordance with the terms of the contract. See 6.3.

4.5 Test Methods and Procedures

4.5.1 Examination for Base Casting Discontinuities - Each base casting shall be visually, penetrant and radiographically examined in accordance with MIL-C-6021. Penetrant examination shall be in accordance with MIL-I-6866 Type I Method A. Evidence of a crack or a cold shut shall be cause for rejection. Radiographic exmaination shall be performed in accordance with MIL-STD-453 and shall consist of six (6) radiographs taken through the base rim (each radiograph focusing on two of the twelve sections of the rim between the ribs), four (4) radiographs focusing on the end of the base rim (taken parallel to the axis of the casting and covering a quadrant of the casting rim with each shot), and at least one radiograph of the floor of the base if sufficient to view the entire floor, otherwise the minimum number required to view the entire floor. The radiographs shall be examined to determine compliance to the discontinuity-severity level requirements specified on the applicable drawing. Any casting failing to meet the visual, penetrant or radiographic requirements shall be classed reject and removed from the lot.

4.5.2 Examination for Mechanical Properties of Base Casting - This test shall be conducted in accordance with MIL-A-21180 on a sample of one base casting out of each heat treat. Two tensile specimens shall be cut from the casting floor and two from the rim

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near sprue gates. The specimens shall then be subjected to a tension test; failure of a single specimen to meet the mechanical properties requirements for the class specified on the drawing shall result in rejection of all the castings of that heat treat. Serialization shall be effected for all castings in the heat, such that each casting is identified with its heat treat.

4.5.3 Air Test Certification of Welds - The welds of the Metal Parts Assembly shall be certified with a test fixture which shall seal the nose end of the assembly while subjecting it to an internal air pressure of 10 pounds per square inch (psi). The specified air pressure shall be applied for a minimum of twenty (20) seconds. Observation shall be made for the escape of air; leakage shall class the unit defective. Leakage shall be determined through the use of a leak test compound which meets the requirements of MIL-L-25567 Type 1. This compound shall be applied to the base-shell, nose-shell, retaining cup, and retaining cup/ring assembly welds.

4.5.4 Metallographic Exmaination of the Electron Beam Weld (Base-Shell) - A test specimen for this weldment shall be manufactured of the identical materials, dimensions, and fit dimensions of the base-shell weldment. A specimen shall be welded at the beginning and end of each shift of each welding machine. The length of the weld shall be a minimum of four inches. The specimen shall be sectioned at its approximate center, and one surface polished and microscopically examined at 20X magnification minimum, (25X is permissible). An acceptable weld shall meet the depth, width, and surface porosity requirements of drawing 9312580 and shall not exhibit evidence of cracking or porosity within .090 inches of the outer surface. At the interface, where cracking cannot be totally eliminated, an acceptable weld shall exhibit a minimum width of .050 inches of weld material containing no cracking. Any specimen failing to meet these requirements shall be classed defective, and the production of that shift shall be rejected and removed from the lot.

5. PREPARATION FOR DELIVERY

5.1 Packing and Marking - Packing and marking shall be in accordance with drawing 9282798.

6. NOTES

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6.1 Intended Use - The components covered by this specification are intended for use on the Patriot Warhead.

6.2 Ordering Data - See MIL-A-48078 (PA).

6.3 Submission of Inspection Equipment Designs For Approval - See MIL-A-48078 - Submit equipment designs, as required, to Commander, US Army Armament Research and Development Command, ATTN: DRDAR-QAR-I.

6.4 Drawings - Drawings listed in Section 2 of this specification under the headings US Army Armament Research and Development Command (ARRADCOM) may also include drawings prepared by, and identified as, Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal or Picatinny Arsenal drawings. Technical Data originally prepared by these activities is now under the cognizance of ARRADCOM.

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Preparing Activity:
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Project Number; 1336-A319

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